B.Tech (1st Semester) **GROUP-C** **Batch (2026-30)**

Course Code- 100107 **Engineering Graphics and Design** **2 0 0 2**

Course Outcomes (COs)	Course Outcome Statements
CO1	Apply standard conventions and methods of engineering drawing for the representation of engineering components.
CO2	Determine true dimensions of constructed points, lines, and planes and using orthographic projections.
CO3	Develop orthographic projection of solids, sectional views of solids for accurate visualization according to engineering applications.
CO4	Draw isometric views, and development of surfaces of a solid object for various engineering applications.
CO5	Develop 2D and 3D drawings using CAD (Computer-Aided Drafting) software for a given engineering drawing and physical object as per industry standards.

Mapped CO(s) Number	Relevant Major Theory Session Outcomes (TSOs)	Relevant Units/ Body of Knowledge (BoK)
CO1	*TSO 1a.* Explain general principles of Engineering Drawing with their significance in engineering communications. *TSO 1b.* Use relevant instruments for engineering drawing. *TSO 1c.* Interpret the given drawing data to solve the given problem. *TSO 1d.* Draw the given regular geometric construction with tangents and normal. *TSO 1e.* Draw the given engineering curve with tangents and normal.	**Unit- 1.0: Fundamentals of Engineering Drawing, Geometrical Constructions and Scales (6 hrs)** 1.1 Principles, significance, BIS conventions, drawing instruments, sheet layout, technical drawings, and dimensioning. 1.2 Drawing Conventions- Different types of lines, lettering, dimensioning systems, symbols. 1.3 Geometric construction related to line, circle, regular Polygons by general method. 1.4 Engineering Scales- Scales on Drawings, Plain scale, Diagonal scale, Vernier scale. 1.5 Engineering Curves- Construction of conic sections (Ellipse, Parabola, and Hyperbola) using General Methods. Cycloidal curves, Involute curves.
CO2	*TSO 2a.* Apply knowledge of First and Third angle method of Projections. *TSO 2b.* Draw 2D projections of given objects *TSO 2c.* Prepare Front View, Top View and Side Views of an object as per first angle projection.	**Unit- 2.0: Orthographic Projection (Multiview Projections) (6 hrs)** 2.1 Principles of Orthographic Projections multi-view projection (First angle and Third angle) conventions. 2.2 Conventions - Projections of Points and lines inclined to both planes

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	TSO 2d. Convert pictorial views into orthographic views. *TSO 2e.* Determine true dimensions of given object as per orthographic projection	2.3 Projections of planes, inclined Planes - Auxiliary Planes. Traces of lines. 2.4 Construction of orthographic projections.
CO3	*TSO 3a.* Draw orthographic Projections of a regular given solid. *TSO 3b.* Develop sectional views of a given object inclined to HP. *TSO 3c.* Develop sectional views of a given object when the sectional plane is inclined to HP. *TSO 3d.* Develop the lateral surfaces of the given regular solids.	**Unit- 3.0: Projection of Regular Solids and Sectional View (6 hrs)** 3.1 Orthographic Projections of regular solids 3.2 Sectional View of Solid when sectional plane is parallel to HP (Horizontal Plane). 3.3 Sectional View of Solid when sectional plane is perpendicular to HP (Horizontal Plane). 3.4 Sectional View of a regular solid when the sectional plane is inclined to HP (Horizontal Plane).
CO4	*TSO 4a.* Apply the concepts of lateral surfaces of given solid (prism/ cylinder/ pyramid/ cone) *TSO 4b.* Use suitable parallel /radial line method in given situations. *TSO 4c.* Apply the concept of sectional solids and prepare development of surfaces. *TSO 4d.* Find out true shape and size of the given sectioned surface. *TSO 4e.* Convert orthographic views into isometric views / projections and vice versa.	**Unit- 4.0: Development of Surfaces & Isometric Projection (5 hrs)** 4.1 Development of lateral surfaces of right regular prisms, cylinders, pyramids and cones resting on HP. 4.2 Parallel Line Method- Used for prisms and cylinders, where all sides or elements are parallel to each other. 4.3 Radial Line Method- Used for pyramids and cones, where the lines or elements radiate outward from a central vertex or apex. 4.4 Isometric Views– conventions – Plane figures, simple and compound solids.
CO5	*TSO 5a.* Develop 2D drawing using CAD software for a given engineering drawing. *TSO 5b.* Develop 2D drawings for physical objects. *TSO 5c.* Develop 3D drawing using CAD software for a given engineering drawing. *TSO 5d.* Develop 3D drawings for physical objects.	**Unit- 5.0: 2D and 3D Drawings Using CAD (Computer-Aided Drafting) Software (5 hrs)** 5.1 2D drawings for a given orthographic projection of objects like screw, nut, bolt etc. 5.2 2D drawings for a given isometric projection of a plumber block. 5.3 2D drawings of physical objects such as screw, nut, bolt, plumber block, washer etc. 5.4 Solid Modeling for a given isometric

Mapped CO(s) Number	Relevant Major Theory Session Outcomes (TSOs)	Relevant Units/ Body of Knowledge (BoK)
		projection. 5.5 Solid Modeling a given physical object.

Reference/Books

S. No.	Titles	Author(s)	Publisher and Edition with ISBN
1.	Engineering Drawing	N. D. Bhatt	Charotar Publishing House
2.	Engineering Drawing	P.J. Shah	S. Chand Publisher
3.	Engineering Drawing	K.Venu Gopal & V.Prabu Raja	New Age Publications
4.	Engineering Drawing and Graphics + AutoCAD	Venugopal K.	New age International Publishers, Delhi, 2022, ISBN: 978-8122472752
5.	Engineering Drawing	Basant Agrawal and Agrawal C. M.	Tata McGraw Hill, New Delhi, 2013 ISBN: 978-1259062889
6.	Engineering Graphics	S. K. Pradhan K.K. Jain	Khanna Book Publishing Company Pvt. Ltd., New Delhi ASIN : B0BM5BMMXT ISBN-10 : 9355381891
7.	Bureau of Indian Standards, Engineering Drawing Practice for Schools and Colleges IS: SP-46, BIS,	BIS, Government of India,	BIS, Government of India, Third Reprint, October 1998; ISBN: 81-7061-091-2.